

CS 2601 Linear and Convex Optimization

12. Newton's method for equality constrained problems

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Outline

- Equality constrained convex QP
- Newton's method with feasible start

Equality constrained convex QP

$$\begin{aligned} \min_{\mathbf{x}} \quad & f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x} + \mathbf{g}^T \mathbf{x} + c \\ \text{s.t.} \quad & \mathbf{A} \mathbf{x} = \mathbf{b} \end{aligned} \tag{QP}$$

where $\mathbf{Q} \in \mathbb{R}^n$, $\mathbf{Q} \succeq \mathbf{O}$, $\mathbf{A} \in \mathbb{R}^{k \times n}$, $\text{rank} \mathbf{A} = k$.

- The Lagrange condition $\nabla \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{0}$ gives the **KKT system** of the problem,

$$\begin{cases} \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{Q} \mathbf{x}^* + \mathbf{g} + \mathbf{A}^T \boldsymbol{\lambda}^* = \mathbf{0} \\ \nabla_{\boldsymbol{\lambda}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{A} \mathbf{x}^* - \mathbf{b} = \mathbf{0} \end{cases} \quad \text{or} \quad \begin{bmatrix} \mathbf{Q} & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix} \begin{bmatrix} \mathbf{x}^* \\ \boldsymbol{\lambda}^* \end{bmatrix} = \begin{bmatrix} -\mathbf{g} \\ \mathbf{b} \end{bmatrix}$$

The coefficient matrix $\mathbf{K} = \begin{bmatrix} \mathbf{Q} & \mathbf{A}^T \\ \mathbf{A} & \mathbf{O} \end{bmatrix}$ is called the **KKT matrix**.

- Any solution to the KKT system gives an optimal solution $\mathbf{x}^*$ with corresponding Lagrange multiplier $\boldsymbol{\lambda}^*$.

Nonsingularity of KKT matrix

Recall $\mathbf{Q} \succeq \mathbf{O}$ and $\text{rank} \mathbf{A} = k$. The following conditions are equivalent

1. $\mathbf{K}$ is nonsingular
2. (QP) has unique minimum
3. $\mathbf{F}^T \mathbf{Q} \mathbf{F} \succ \mathbf{O}$ for any $\mathbf{F} \in \mathbb{R}^{n \times (n-k)}$ s.t. $\text{Range}(\mathbf{F}) = \text{Null}(\mathbf{A})$, i.e. the columns of $\mathbf{F}$ form a basis of $\text{Null}(\mathbf{A})$.
4. $\mathbf{A} \mathbf{v} = \mathbf{0}, \mathbf{v} \neq \mathbf{0} \implies \mathbf{v}^T \mathbf{Q} \mathbf{v} > 0$, i.e. $\mathbf{Q}$ is positive definite on the nullspace of $\mathbf{A}$.
5. $\text{Null}(\mathbf{Q}) \cap \text{Null}(\mathbf{A}) = \{\mathbf{0}\}$, i.e. $\mathbf{Q}$ and $\mathbf{A}$ have no nontrivial common nullspace, i.e. $\mathbf{A} \mathbf{v} = \mathbf{0}, \mathbf{Q} \mathbf{v} = \mathbf{0}$ only have the trivial solution $\mathbf{v} = \mathbf{0}$.

In particular, if $\mathbf{Q} \succ \mathbf{O}$, then $\mathbf{K}$ is nonsingular (by 4).

Proof

We show $1 \iff 2 \iff 3 \iff 4$.

- $(1 \iff 2)$. Since A has full row rank, each $\mathbf{x}^*$ corresponds to a unique $\boldsymbol{\lambda}^*$ satisfying

$$\mathbf{Q}\mathbf{x}^* + \mathbf{g} + \mathbf{A}^T \boldsymbol{\lambda}^* = \mathbf{0}.$$

Thus # of minima of (QP) is equal to # of solutions to the KKT system. Hence $1 \iff$ KKT system has a unique solution $\iff 2$.

- $(2 \iff 3)$. Given a feasible $\mathbf{x}_0$, (QP) is equivalent to

$$\min \quad g(\mathbf{z}) = f(\mathbf{x}_0 + \mathbf{F}\mathbf{z}) = f(\mathbf{z}_0) + (\mathbf{Q}\mathbf{x}_0 + \mathbf{g})^T \mathbf{F}\mathbf{z} + \frac{1}{2} \mathbf{z}^T \mathbf{F}^T \mathbf{Q} \mathbf{F} \mathbf{z}$$

which has a unique minimum iff $\mathbf{F}^T \mathbf{Q} \mathbf{F} \succ \mathbf{O}$ (slide 8 of §5).

The minima of $g(\mathbf{z})$ and (QP) are in one-to-one correspondence, so $2 \iff 3$.

Proof (cont'd)

- $(3 \Leftrightarrow 4)$. Note $\mathbf{v} \in \text{Null}(\mathbf{A})$ iff $\mathbf{v} = \mathbf{Fz}$, and $\mathbf{Fz} \neq \mathbf{0}$ iff $\mathbf{z} \neq \mathbf{0}$,

$$3 \iff (\mathbf{z}^T \mathbf{F}^T \mathbf{Q} \mathbf{F} \mathbf{z} > 0 \text{ if } \mathbf{z} \neq \mathbf{0})$$

$$\iff (\mathbf{v}^T \mathbf{Q} \mathbf{v} > 0 \text{ if } \mathbf{v} = \mathbf{Fz}, \mathbf{z} \neq \mathbf{0})$$

$$\iff (\mathbf{v}^T \mathbf{Q} \mathbf{v} > 0 \text{ if } \mathbf{A}\mathbf{v} = \mathbf{0}, \mathbf{v} \neq \mathbf{0})$$

- $(4 \Leftrightarrow 5)$. Since $\mathbf{Q} \succeq \mathbf{O}$, $\mathbf{v}^T \mathbf{Q} \mathbf{v} = 0$ iff $\mathbf{Q} \mathbf{v} = \mathbf{0}$ (see the Lemma below).

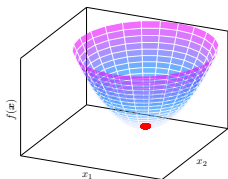
$$4 \iff (\mathbf{A}\mathbf{v} = \mathbf{0} \text{ and } \mathbf{v} \neq \mathbf{0} \implies \mathbf{Q}\mathbf{v} \neq \mathbf{0})$$

$$\iff (\mathbf{A}\mathbf{v} = \mathbf{0} \text{ and } \mathbf{Q}\mathbf{v} = \mathbf{0} \implies \mathbf{v} = \mathbf{0}) \iff 5$$

Lemma. Suppose $\mathbf{Q} \succeq \mathbf{O}$. Then $\mathbf{v}^T \mathbf{Q} \mathbf{v} = 0 \iff \mathbf{Q}\mathbf{v} = \mathbf{0}$.

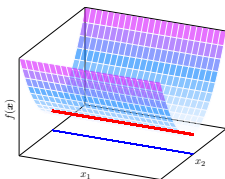
Proof. Let $\mathbf{Q} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$. Then $\mathbf{Q} = \mathbf{B}^T \mathbf{B}$ with $\mathbf{B} = \mathbf{\Lambda}^{\frac{1}{2}} \mathbf{U}^T$. Thus $\mathbf{v}^T \mathbf{Q} \mathbf{v} = \|\mathbf{B}\mathbf{v}\|^2 = 0 \implies \mathbf{B}\mathbf{v} = \mathbf{0} \implies \mathbf{Q}\mathbf{v} = \mathbf{B}^T \mathbf{B}\mathbf{v} = \mathbf{0}$. The other direction is trivial.

Unconstrained vs constrained problems



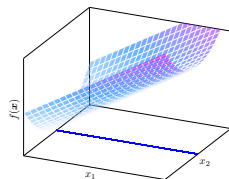
$$Q \succ 0$$

unique solution



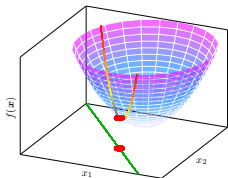
$$g \perp \text{Null}(Q) \neq \{0\}$$

infinitely many solutions



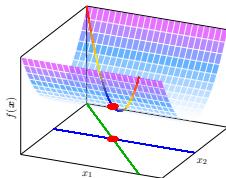
$$g \not\perp \text{Null}(Q) \neq \{0\}$$

no solution



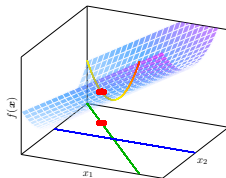
$$Q \succ 0$$

unique solution



$$\text{Null}(Q) \cap \text{Null}(A) = \{0\}$$

unique solution



$$\text{Null}(Q) \cap \text{Null}(A) = \{0\}$$

unique solution

Example

$$\begin{array}{ll}\min_{x_1, x_2} & f(x_1, x_2) = \frac{1}{2}x_2^2 \\ \text{s.t.} & x_1 + 2x_2 = b\end{array}$$

Trivial with solution $x_1^* = b, x_2^* = 0$.

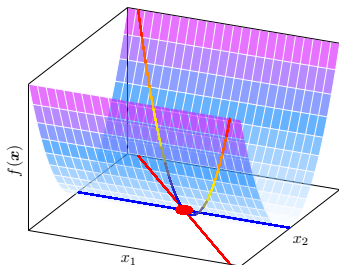
Let's check Condition 3 on slide 3.

$$\mathbf{Q} = \text{diag}\{0, 1\}, \quad \mathbf{A} = [1, 2]$$

Let $\mathbf{F} = [2, -1]^T$. Then $\text{Range}(\mathbf{F}) = \text{Null}(\mathbf{A})$, and

$$\mathbf{F}^T \mathbf{Q} \mathbf{F} = [1] \succ \mathbf{0}$$

Note. The unconstrained problem $\min_{\mathbf{x}} f(\mathbf{x})$ has infinitely many solutions. But this does not prevent the constrained problem from having a unique solution, as $\mathbf{Q} \succ \mathbf{0}$ on $\text{Null}(\mathbf{A})$ (see item 4 on slide 3).



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Newton direction

Consider the convex problem with equality constraints,

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{Ax} = \mathbf{b}\end{array}$$

The **Newton direction** at $\mathbf{x}_k$ is the solution to the approximation

$$\begin{array}{ll}\min_{\mathbf{v}} & \hat{f}(\mathbf{x}_k + \mathbf{v}) = f(\mathbf{x}_k) + \nabla f(\mathbf{x}_k)^T \mathbf{v} + \frac{1}{2} \mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v} \\ \text{s.t.} & \mathbf{A}(\mathbf{x}_k + \mathbf{v}) = \mathbf{b}\end{array}$$

For **feasible** $\mathbf{x}_k$, the constraints become $\mathbf{Av} = \mathbf{0}$. The KKT system is

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}_k) & \mathbf{A}^T \\ \mathbf{A} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}_k) \\ \mathbf{0} \end{bmatrix}$$

We will assume the KKT matrix is always nonsingular (cf. slide 3).

Newton direction (cont'd)

The Newton direction is a descent direction when $\mathbf{v} \neq \mathbf{0}$. In fact,

$$\left. \frac{d}{dt} f(\mathbf{x}_k + t\mathbf{v}) \right|_{t=0} = \nabla f(\mathbf{x}_k)^T \mathbf{v}$$

by the first KKT equation,

$$\begin{aligned} &= -(\nabla^2 f(\mathbf{x}_k) \mathbf{v} + \mathbf{A}^T \boldsymbol{\lambda})^T \mathbf{v} \\ &= -\mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v} - \boldsymbol{\lambda}^T \mathbf{A} \mathbf{v} \end{aligned}$$

by the second KKT equation,

$$= -\mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v}$$

Since $\mathbf{v} \in \text{Null}(\mathbf{A})$, item 4 of slide 3 implies $\mathbf{v}^T \nabla^2 f(\mathbf{x}_k) \mathbf{v} > 0$ and hence

$$\left. \frac{d}{dt} f(\mathbf{x}_k + t\mathbf{v}) \right|_{t=0} < 0$$

Linearization of Lagrange condition

The Lagrange condition for the original problem is

$$\begin{aligned}\nabla f(\mathbf{x}^*) + \mathbf{A}^T \boldsymbol{\lambda} &= \mathbf{0} \\ \mathbf{A} \mathbf{x}^* &= \mathbf{b}\end{aligned}$$

Let $\mathbf{x}^* = \mathbf{x}_k + \mathbf{v}$ for a feasible $\mathbf{x}_k$. The above equations become

$$\begin{aligned}\nabla f(\mathbf{x}_k + \mathbf{v}) + \mathbf{A}^T \boldsymbol{\lambda} &= \mathbf{0} \\ \mathbf{A} \mathbf{v} &= \mathbf{0}\end{aligned}$$

If we linearize the first equation by $\nabla f(\mathbf{x}_k + \mathbf{v}) \approx \nabla f(\mathbf{x}_k) + \nabla^2 f(\mathbf{x}_k) \mathbf{v}$, then

$$\begin{aligned}\nabla f(\mathbf{x}_k) + \nabla^2 f(\mathbf{x}_k) \mathbf{v} + \mathbf{A}^T \boldsymbol{\lambda} &\approx \mathbf{0} \\ \mathbf{A} \mathbf{v} &= \mathbf{0}\end{aligned}$$

Newton's method can be considered an iterative method for solving the Lagrange condition.

(Damped) Newton's method with feasible start

1: initialization $\mathbf{x} \leftarrow \mathbf{x}_0 \in X$ $\triangleright \mathbf{x}_0$ is feasible, i.e. $A\mathbf{x}_0 = \mathbf{b}$

2: **repeat**

3: **Compute Newton's direction $\mathbf{v}$ by solving**

$$\begin{bmatrix} \nabla^2 f(\mathbf{x}) & A^T \\ A & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{v} \\ \lambda \end{bmatrix} = \begin{bmatrix} -\nabla f(\mathbf{x}) \\ \mathbf{0} \end{bmatrix}$$

4: $t \leftarrow 1$ $\triangleright$ backtracking line search on lines 4-7

5: **while** $f(\mathbf{x} + t\mathbf{v}) > f(\mathbf{x}) + \alpha t \nabla f(\mathbf{x})^T \mathbf{v}$ **do**

6: $t \leftarrow \beta t$

7: **end while**

8: $\mathbf{x} \leftarrow \mathbf{x} + t\mathbf{v}$

9: **until** $\|\mathbf{v}\| \leq \delta$

10: **return** $\mathbf{x}$

Note. We **cannot** use $\|\nabla f(\mathbf{x})\| \leq \delta$ as stopping criterion now, as $\nabla f(\mathbf{x}^*) = \mathbf{0}$ no longer holds in general. [BV] uses $\sqrt{\mathbf{v}^T \nabla^2 f(\mathbf{x}) \mathbf{v}} \leq \delta$.

Note. This is called a **feasible descent method**, since all $\mathbf{x}_k$ are feasible and $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$ unless $\mathbf{x}_k$ is optimal.

Convergence analysis

Let $\mathbf{F} \in \mathbb{R}^{n \times (n-k)}$ be a matrix whose columns are linearly independent solutions to $\mathbf{A}\mathbf{v} = \mathbf{0}$. Fix a feasible $\tilde{\mathbf{x}} \in X$. Every $\mathbf{x} \in X$ has a unique representation of the form $\mathbf{x} = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}$,

$$X = \{\mathbf{x} : \mathbf{A}\mathbf{x} = \mathbf{b}\} = \{\tilde{\mathbf{x}} + \mathbf{F}\mathbf{z} : \mathbf{z} \in \mathbb{R}^{n-k}\}$$

Constrained problem reduces to unconstrained problem by $\mathbf{x} = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}$,

$$(C) : \begin{cases} \min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s.t.} & \mathbf{A}\mathbf{x} = \mathbf{b} \end{cases} \iff (U) : \min_{\mathbf{z}} g(\mathbf{z}) = f(\tilde{\mathbf{x}} + \mathbf{F}\mathbf{z})$$

Applying Newton's method to (C) with initial point $\mathbf{x}_0 = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_0$ is equivalent to applying Newton's method to (U) with initial point $\mathbf{z}_0$:

If $\{\mathbf{x}_k\}$ and $\{\mathbf{z}_k\}$ are the iterates for (C) and (U), respectively, then

$$\mathbf{x}_k = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_k, \quad \forall k$$

Newton's method has same convergence properties for (C) and (U). 13

Proof

We only need to show $\mathbf{x}_1 = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_1$ and then use induction.

The Newton direction $\mathbf{v}_0$ at $\mathbf{x}_0$ for (C) is the unique solution to

$$\begin{aligned} \min_{\mathbf{v}} \quad & \hat{f}(\mathbf{x}_0 + \mathbf{v}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T \mathbf{v} + \frac{1}{2} \mathbf{v}^T \nabla^2 f(\mathbf{x}_0) \mathbf{v} \\ \text{s.t.} \quad & \mathbf{A}\mathbf{v} = \mathbf{0} \end{aligned}$$

The Newton direction $\mathbf{u}_0$ at $\mathbf{z}_0$ for (U) is the unique (why?) solution to

$$\begin{aligned} \min_{\mathbf{u}} \quad & \hat{g}(\mathbf{z}_0 + \mathbf{u}) = g(\mathbf{z}_0) + \nabla g(\mathbf{z}_0)^T \mathbf{u} + \frac{1}{2} \mathbf{u}^T \nabla^2 g(\mathbf{z}_0) \mathbf{u} \\ & = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T \mathbf{F}\mathbf{u} + \frac{1}{2} \mathbf{u}^T \mathbf{F}^T \nabla^2 f(\mathbf{x}_0) \mathbf{F}\mathbf{u} \end{aligned}$$

where, by the chain rule, $\nabla g(\mathbf{z}_0) = \mathbf{F}^T \nabla f(\mathbf{x}_0)$, $\nabla^2 g(\mathbf{z}_0) = \mathbf{F}^T \nabla^2 f(\mathbf{x}_0) \mathbf{F}$.

The two problems are related by $\mathbf{v} = \mathbf{F}\mathbf{u}$. In particular, $\mathbf{v}_0 = \mathbf{F}\mathbf{u}_0$.

Proof (cont'd)

Now we show backtracking line search gives the same step size t_0 for both (C) and (U).

Note

$$f(\mathbf{x}_0) = f(\tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_0) = g(\mathbf{z}_0)$$

$$f(\mathbf{x}_0 + t\mathbf{v}_0) = f(\tilde{\mathbf{x}} + \mathbf{F}(\mathbf{z}_0 + t\mathbf{u}_0)) = g(\mathbf{z}_0 + t\mathbf{u}_0)$$

$$\nabla f(\mathbf{x}_0)^T \mathbf{v}_0 = \nabla f(\mathbf{x}_0)^T \mathbf{F}\mathbf{u}_0 = \nabla g(\mathbf{z}_0)^T \mathbf{u}_0$$

Thus the test condition in backtracking line search for f

$$f(\mathbf{x}_0 + t\mathbf{v}_0) > f(\mathbf{x}_0) + \alpha t \nabla f(\mathbf{x}_0)^T \mathbf{v}_0$$

is exactly the same as that for g ,

$$g(\mathbf{z}_0 + t\mathbf{u}_0) > g(\mathbf{z}_0) + \alpha t \nabla g(\mathbf{z}_0)^T \mathbf{u}_0$$

Finally,

$$\mathbf{x}_1 = \mathbf{x}_0 + t_0\mathbf{v}_0 = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_0 + t_0\mathbf{F}\mathbf{u}_0 = \tilde{\mathbf{x}} + \mathbf{F}(\mathbf{z}_0 + t_0\mathbf{u}_0) = \tilde{\mathbf{x}} + \mathbf{F}\mathbf{z}_1$$